LTSpice Notch Quality Report

CROSS COUPLED3

Auto-detected swept parameters and quality-factor (Q = f0 / BW @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 1

Total simulation steps : 88

`V(act+,act-)`

swept parameters: Cp, Itail

Report contents (per signal column):

1. All curves — Bode plot (magnitude + phase)
2. Magnitude grouped by each swept parameter (one subplot per fixed value)
3. Phase grouped by each swept parameter (one subplot per fixed value)
4. Quality-factor table for that column

End of report (extra analyses):

A. Cross-signal comparison (act vs pas, if both present)

B. Q vs each swept parameter (one line per other-param combo)

C. BW vs each swept parameter

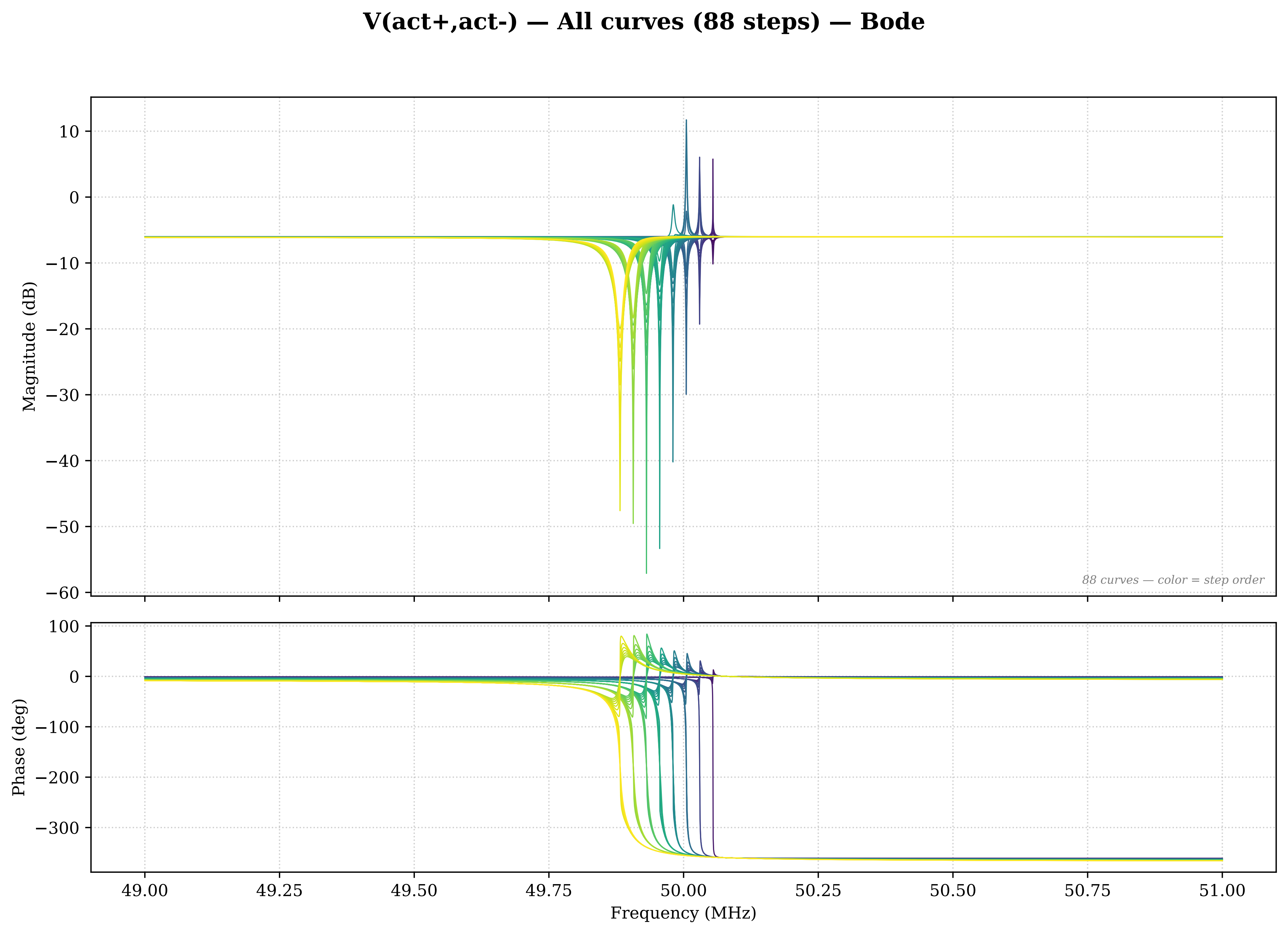
D. Q heatmap for V(act): rows × columns swept params, colored

E. Combined quality-factor table across all signals

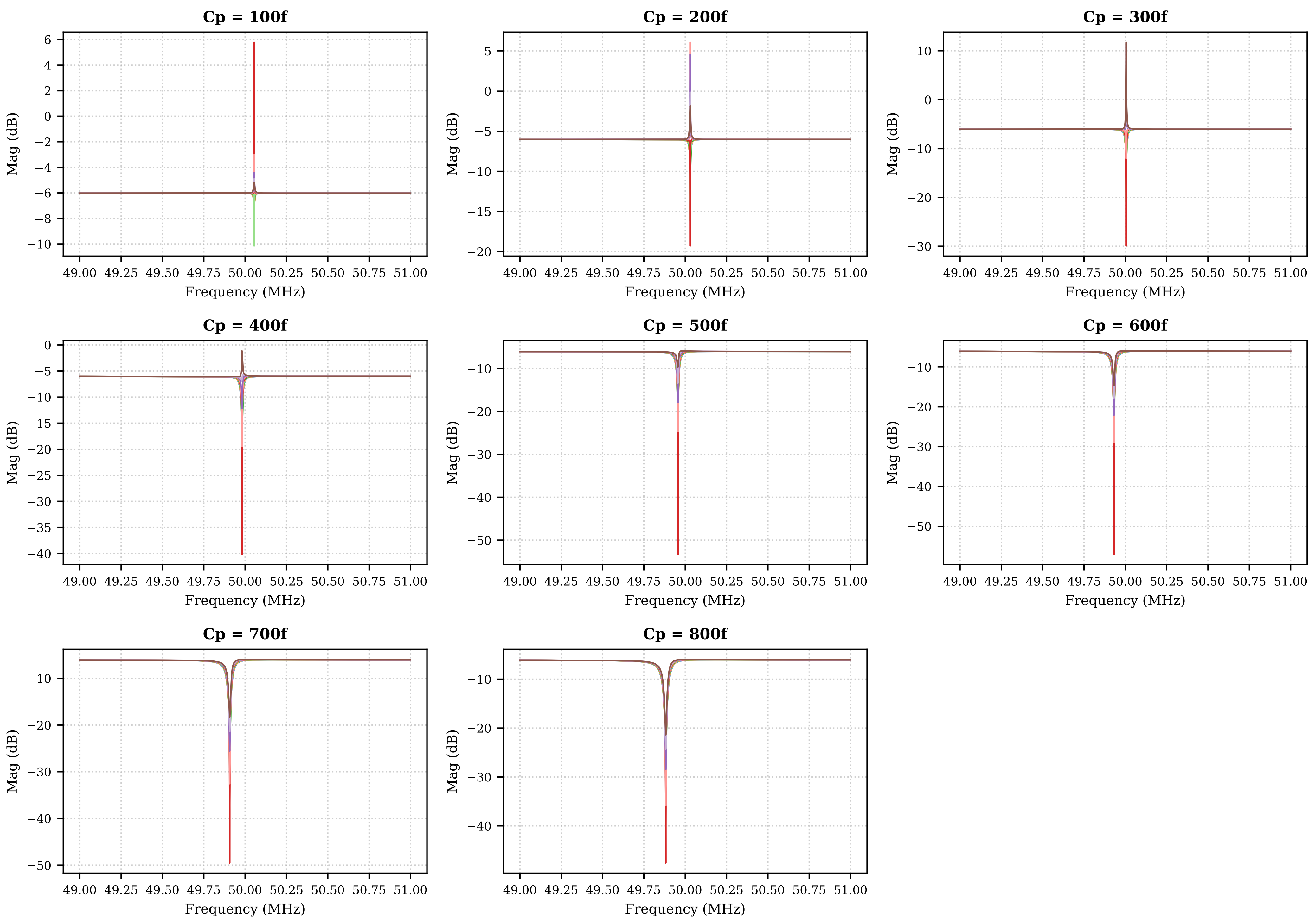
Signal: V(act+,act-)

Swept parameters: Cp, Itail

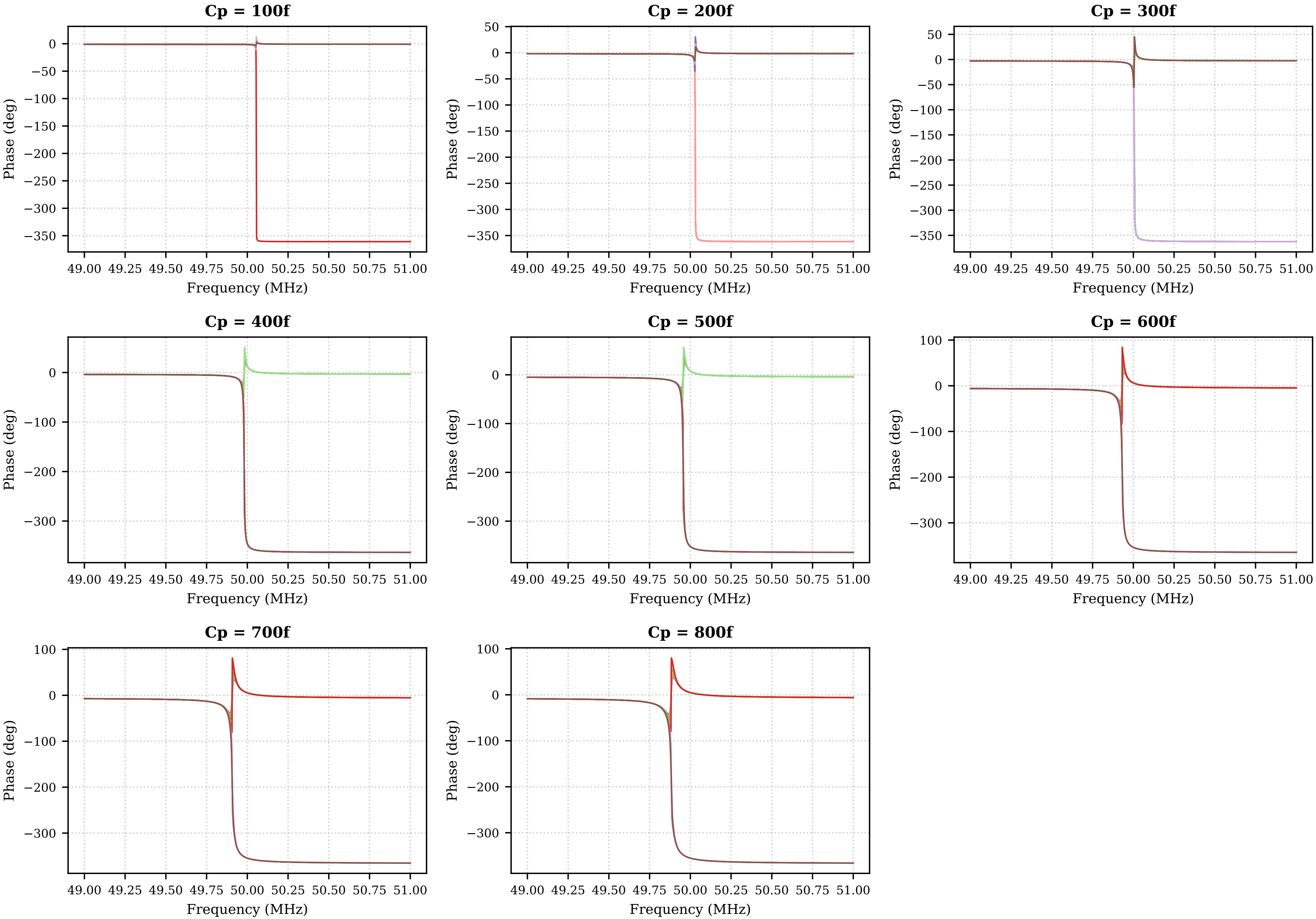
Steps: 88


V(act+,act-) — All curves (88 steps) — Bode
Magnitude (dB)
10
0
−10
−20
−30
−40
−50
−60
88 curves — color = step order
Phase (deg)
100
0
−100
−200
−300
49.00
49.25
49.50
49.75
50.00
50.25
50.50
50.75
51.00
Frequency (MHz)

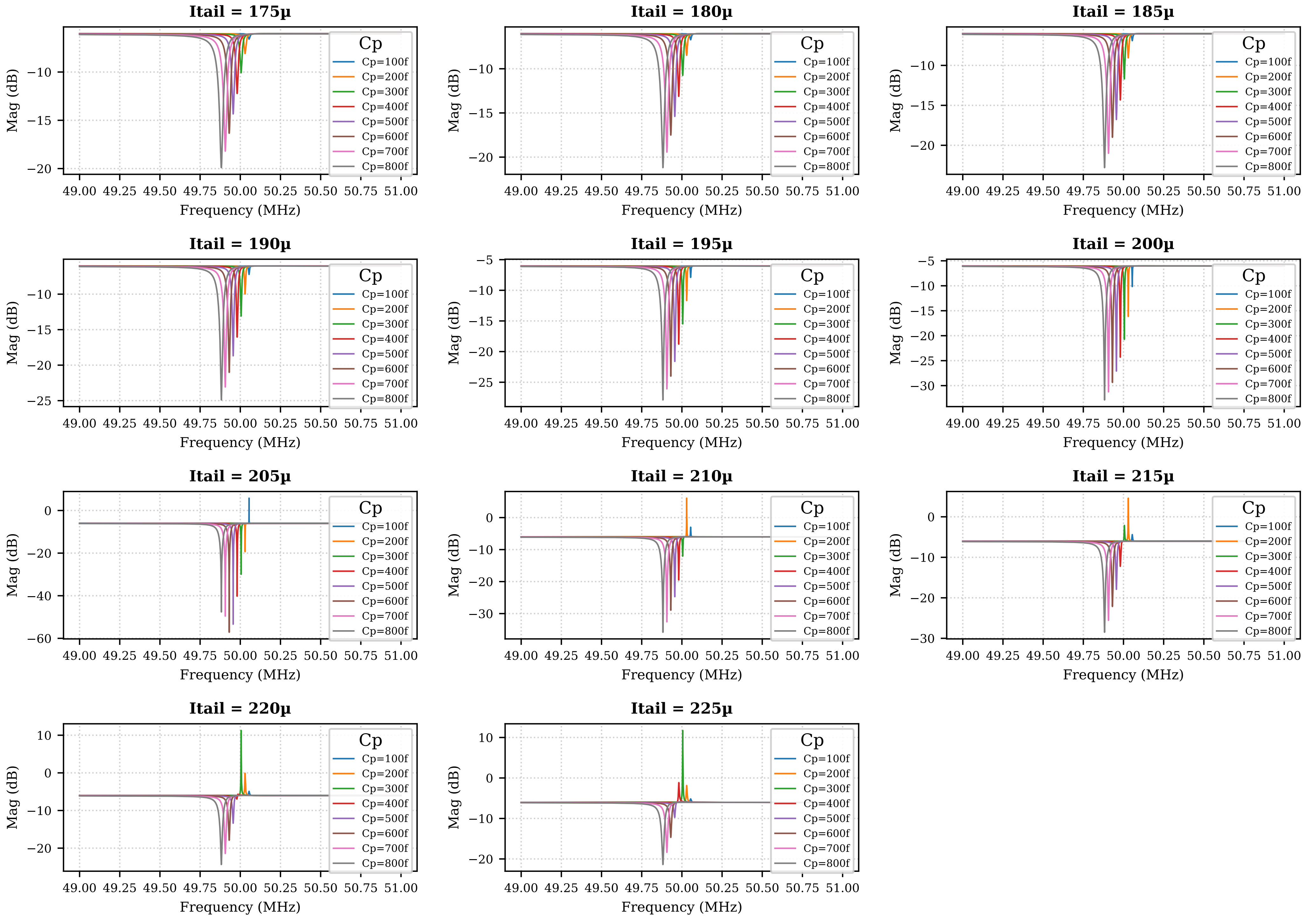
V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)



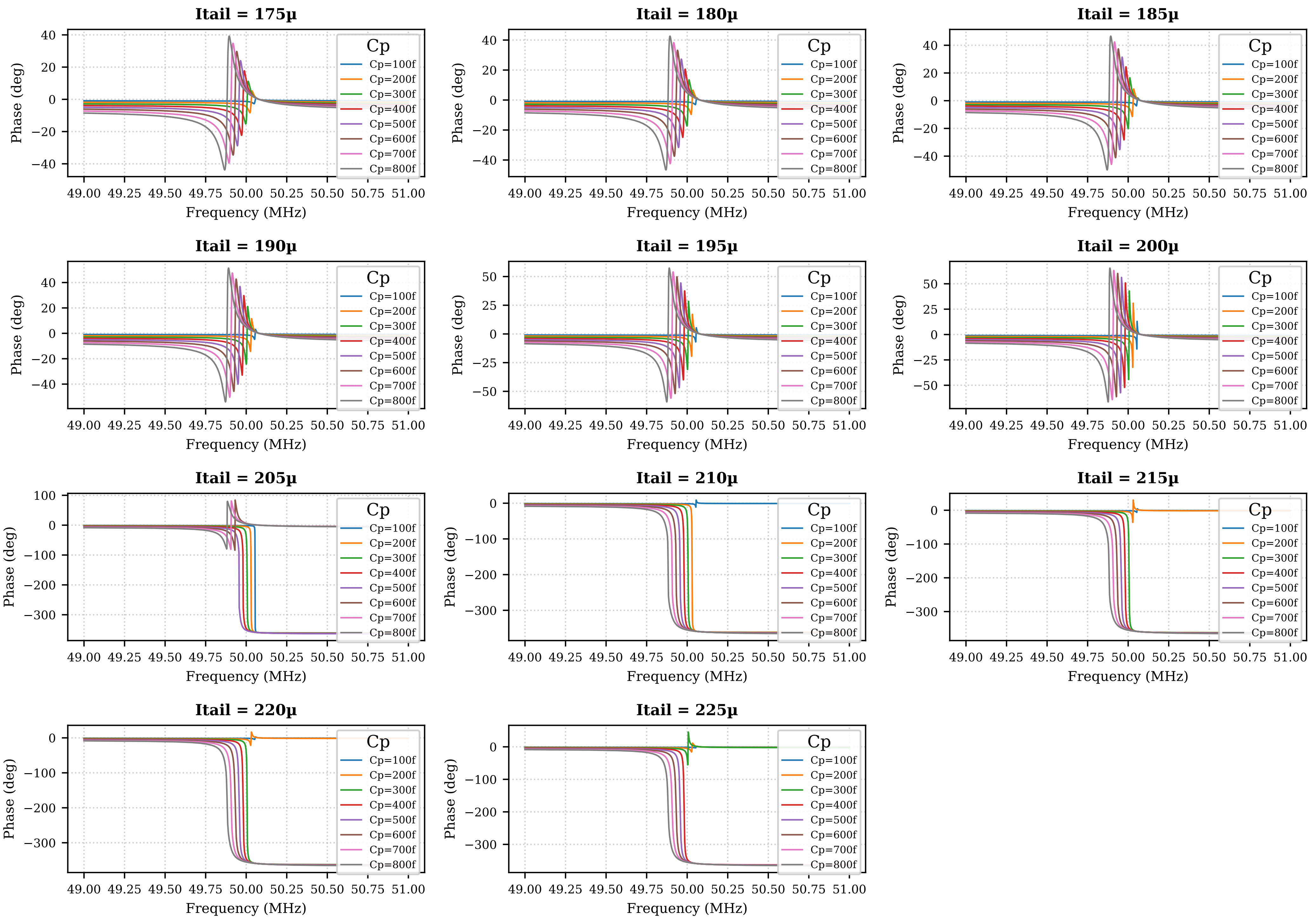
V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)



V(act+,act-) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)



V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)



V(act+,act-) — Quality factor table (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	175µ	50.05	-6.022	0.5711			notch shallower than 3 d
V(act+,act-)	100f	180µ	50.05	-6.022	0.6939			notch shallower than 3 d
V(act+,act-)	100f	185µ	50.05	-6.022	0.8808			notch shallower than 3 d
V(act+,act-)	100f	190µ	50.05	-6.022	1.199			notch shallower than 3 d
V(act+,act-)	100f	195µ	50.05	-6.022	1.865			notch shallower than 3 d
V(act+,act-)	100f	200µ	50.05	-6.022	4.135	999.6	5.007e+04	
V(act+,act-)	100f	205µ	50.04	-6.022	0.001893			notch shallower than 3 d
V(act+,act-)	100f	210µ	49.94	-6.022	0.0002734			notch shallower than 3 d
V(act+,act-)	100f	215µ	49.84	-6.022	0.0001498			notch shallower than 3 d
V(act+,act-)	100f	220µ	49.75	-6.022	0.000104			notch shallower than 3 d
V(act+,act-)	100f	225µ	49.68	-6.022	8.006e-05			notch shallower than 3 d
V(act+,act-)	200f	175µ	50.03	-6.025	2.06			notch shallower than 3 d
V(act+,act-)	200f	180µ	50.03	-6.025	2.455			notch shallower than 3 d
V(act+,act-)	200f	185µ	50.03	-6.025	3.029	860.4	5.815e+04	
V(act+,act-)	200f	190µ	50.03	-6.025	3.943	3771	1.327e+04	
V(act+,act-)	200f	195µ	50.03	-6.025	5.643	4404	1.136e+04	
V(act+,act-)	200f	200µ	50.03	-6.025	10.11	4071	1.229e+04	
V(act+,act-)	200f	205µ	50.03	-6.025	13.27	2456	2.037e+04	
V(act+,act-)	200f	210µ	50	-6.025	0.00865			notch shallower than 3 d
V(act+,act-)	200f	215µ	49.95	-6.025	0.002867			notch shallower than 3 d
V(act+,act-)	200f	220µ	49.89	-6.025	0.001741			notch shallower than 3 d
V(act+,act-)	200f	225µ	49.84	-6.025	0.001264			notch shallower than 3 d
V(act+,act-)	300f	175µ	50	-6.03	4.046	8680	5761	
V(act+,act-)	300f	180µ	50	-6.03	4.72	9532	5246	
V(act+,act-)	300f	185µ	50	-6.03	5.658	9877	5063	
V(act+,act-)	300f	190µ	50	-6.03	7.06	9789	5108	
V(act+,act-)	300f	195µ	50	-6.03	9.436	9277	5390	
V(act+,act-)	300f	200µ	50	-6.03	14.7	8273	6044	
V(act+,act-)	300f	205µ	50	-6.03	23.91	6581	7598	
V(act+,act-)	300f	210µ	50	-6.03	6.015	3316	1.508e+04	
V(act+,act-)	300f	215µ	49.99	-6.03	0.04976			notch shallower than 3 d
V(act+,act-)	300f	220µ	49.95	-6.03	0.01447			notch shallower than 3 d

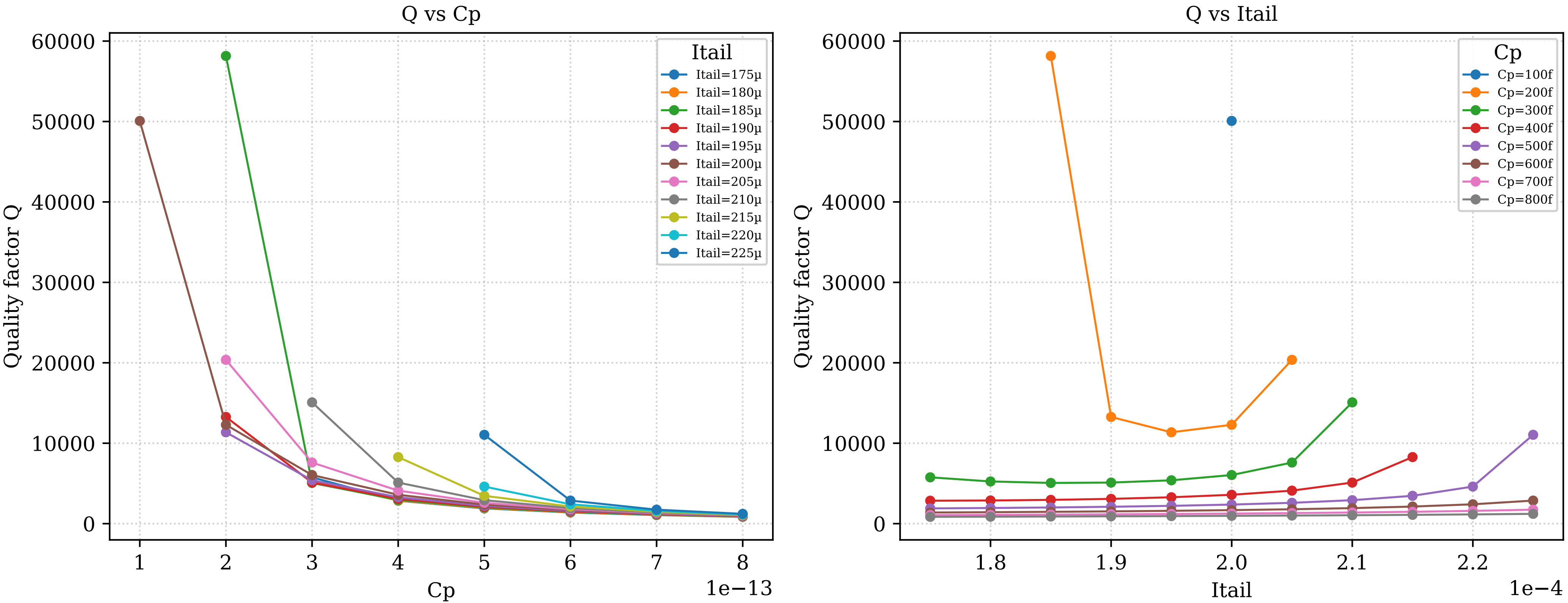
V(act+,act-) — Quality factor table (page 2/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	300f	225µ	49.92	-6.03	0.008533			notch shallower than 3 d
V(act+,act-)	400f	175µ	49.98	-6.038	6.186	1.753e+04	2851	
V(act+,act-)	400f	180µ	49.98	-6.038	7.082	1.735e+04	2881	
V(act+,act-)	400f	185µ	49.98	-6.038	8.285	1.691e+04	2955	
V(act+,act-)	400f	190µ	49.98	-6.038	10.01	1.622e+04	3082	
V(act+,act-)	400f	195µ	49.98	-6.038	12.75	1.524e+04	3280	
V(act+,act-)	400f	200µ	49.98	-6.038	18.26	1.392e+04	3590	
V(act+,act-)	400f	205µ	49.98	-6.038	34.16	1.218e+04	4104	
V(act+,act-)	400f	210µ	49.98	-6.038	13.45	9786	5107	
V(act+,act-)	400f	215µ	49.98	-6.038	6.175	6039	8277	
V(act+,act-)	400f	220µ	49.98	-6.038	0.9208			notch shallower than 3 d
V(act+,act-)	400f	225µ	49.95	-6.038	0.06877			notch shallower than 3 d
V(act+,act-)	500f	175µ	49.96	-6.048	8.292	2.634e+04	1897	
V(act+,act-)	500f	180µ	49.96	-6.048	9.346	2.567e+04	1946	
V(act+,act-)	500f	185µ	49.96	-6.048	10.73	2.483e+04	2012	
V(act+,act-)	500f	190µ	49.96	-6.048	12.64	2.38e+04	2099	
V(act+,act-)	500f	195µ	49.96	-6.048	15.55	2.255e+04	2215	
V(act+,act-)	500f	200µ	49.96	-6.048	21.03	2.106e+04	2372	
V(act+,act-)	500f	205µ	49.96	-6.048	47.3	1.928e+04	2591	
V(act+,act-)	500f	210µ	49.96	-6.047	18.67	1.713e+04	2917	
V(act+,act-)	500f	215µ	49.96	-6.047	11.86	1.444e+04	3460	
V(act+,act-)	500f	220µ	49.96	-6.047	7.332	1.084e+04	4610	
V(act+,act-)	500f	225µ	49.96	-6.047	3.676	4522	1.105e+04	
V(act+,act-)	600f	175µ	49.93	-6.061	10.28	3.602e+04	1386	
V(act+,act-)	600f	180µ	49.93	-6.06	11.44	3.506e+04	1424	
V(act+,act-)	600f	185µ	49.93	-6.06	12.94	3.395e+04	1471	
V(act+,act-)	600f	190µ	49.93	-6.06	14.95	3.27e+04	1527	
V(act+,act-)	600f	195µ	49.93	-6.06	17.94	3.129e+04	1596	
V(act+,act-)	600f	200µ	49.93	-6.06	23.29	2.969e+04	1682	
V(act+,act-)	600f	205µ	49.93	-6.06	51.05	2.789e+04	1790	
V(act+,act-)	600f	210µ	49.93	-6.06	22.88	2.585e+04	1932	
V(act+,act-)	600f	215µ	49.93	-6.06	16.07	2.35e+04	2125	

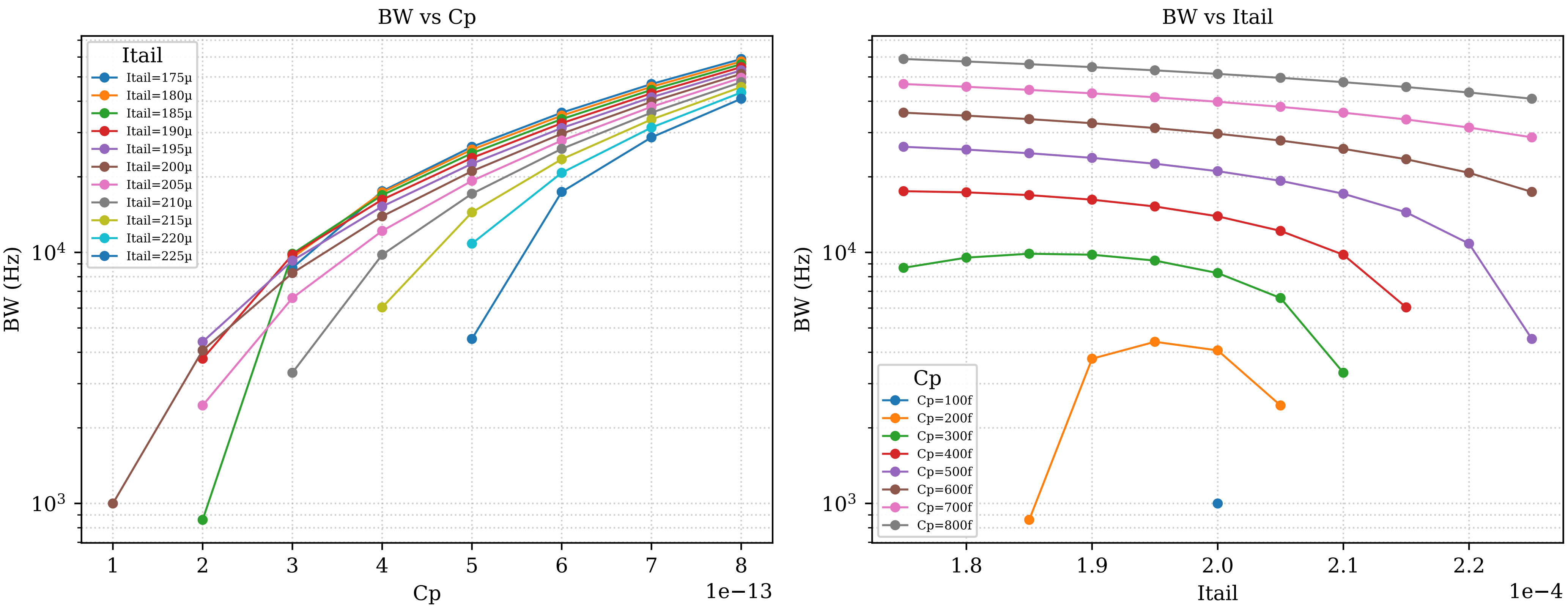
V(act+,act-) — Quality factor table (page 3/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	600f	220µ	49.93	-6.06	11.83	2.075e+04	2406	
V(act+,act-)	600f	225µ	49.93	-6.059	8.604	1.741e+04	2867	
V(act+,act-)	700f	175µ	49.91	-6.076	12.12	4.685e+04	1065	
V(act+,act-)	700f	180µ	49.91	-6.076	13.36	4.568e+04	1093	
V(act+,act-)	700f	185µ	49.91	-6.075	14.92	4.439e+04	1124	
V(act+,act-)	700f	190µ	49.91	-6.075	17	4.3e+04	1161	
V(act+,act-)	700f	195µ	49.91	-6.075	20	4.147e+04	1203	
V(act+,act-)	700f	200µ	49.91	-6.075	25.17	3.981e+04	1254	
V(act+,act-)	700f	205µ	49.91	-6.075	43.47	3.8e+04	1313	
V(act+,act-)	700f	210µ	49.91	-6.075	26.49	3.602e+04	1385	
V(act+,act-)	700f	215µ	49.91	-6.075	19.49	3.384e+04	1475	
V(act+,act-)	700f	220µ	49.91	-6.074	15.34	3.143e+04	1588	
V(act+,act-)	700f	225µ	49.91	-6.074	12.29	2.872e+04	1738	
V(act+,act-)	800f	175µ	49.88	-6.094	13.81	5.893e+04	846.5	
V(act+,act-)	800f	180µ	49.88	-6.094	15.1	5.761e+04	865.9	
V(act+,act-)	800f	185µ	49.88	-6.094	16.7	5.62e+04	887.6	
V(act+,act-)	800f	190µ	49.88	-6.093	18.8	5.47e+04	911.9	
V(act+,act-)	800f	195µ	49.88	-6.093	21.79	5.311e+04	939.3	
V(act+,act-)	800f	200µ	49.88	-6.093	26.77	5.14e+04	970.4	
V(act+,act-)	800f	205µ	49.88	-6.093	41.51	4.959e+04	1006	
V(act+,act-)	800f	210µ	49.88	-6.092	29.72	4.765e+04	1047	
V(act+,act-)	800f	215µ	49.88	-6.092	22.39	4.557e+04	1095	
V(act+,act-)	800f	220µ	49.88	-6.092	18.25	4.334e+04	1151	
V(act+,act-)	800f	225µ	49.88	-6.092	15.28	4.093e+04	1219	

B. V(act+,act-) — Quality factor vs swept parameters



C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=175µ	Itail=180µ	Itail=185µ	Itail=190µ	Itail=195µ	Itail=200µ	Itail=205µ	Itail=210µ	Itail=215µ	Itail=220µ	Itail=225µ
Cp=100f	no BW	no BW	no BW	no BW	no BW	50075	no BW	no BW	no BW	no BW	no BW
Cp=200f	no BW	no BW	58150	13266	11359	12291	20367	no BW	no BW	no BW	no BW
Cp=300f	5761	5246	5063	5108	5390	6044	7598	15079	no BW	no BW	no BW
Cp=400f	2851	2881	2955	3082	3280	3590	4104	5107	8277	no BW	no BW
Cp=500f	1897	1946	2012	2099	2215	2372	2591	2917	3460	4610	11047
Cp=600f	1386	1424	1471	1527	1596	1682	1790	1932	2125	2406	2867
Cp=700f	1065	1093	1124	1161	1203	1254	1313	1385	1475	1588	1738
Cp=800f	847	866	888	912	939	970	1006	1047	1095	1151	1219

Red = BW could not be measured | Yellow = Q computed (Q ≤ 1400) | Green = Q > 1400

E. Combined quality-factor table — all signals (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	175µ	50.05	-6.022	0.5711			notch shallower than 3 d
V(act+,act-)	100f	180µ	50.05	-6.022	0.6939			notch shallower than 3 d
V(act+,act-)	100f	185µ	50.05	-6.022	0.8808			notch shallower than 3 d
V(act+,act-)	100f	190µ	50.05	-6.022	1.199			notch shallower than 3 d
V(act+,act-)	100f	195µ	50.05	-6.022	1.865			notch shallower than 3 d
V(act+,act-)	100f	200µ	50.05	-6.022	4.135	999.6	5.007e+04	
V(act+,act-)	100f	205µ	50.04	-6.022	0.001893			notch shallower than 3 d
V(act+,act-)	100f	210µ	49.94	-6.022	0.0002734			notch shallower than 3 d
V(act+,act-)	100f	215µ	49.84	-6.022	0.0001498			notch shallower than 3 d
V(act+,act-)	100f	220µ	49.75	-6.022	0.000104			notch shallower than 3 d
V(act+,act-)	100f	225µ	49.68	-6.022	8.006e-05			notch shallower than 3 d
V(act+,act-)	200f	175µ	50.03	-6.025	2.06			notch shallower than 3 d
V(act+,act-)	200f	180µ	50.03	-6.025	2.455			notch shallower than 3 d
V(act+,act-)	200f	185µ	50.03	-6.025	3.029	860.4	5.815e+04	
V(act+,act-)	200f	190µ	50.03	-6.025	3.943	3771	1.327e+04	
V(act+,act-)	200f	195µ	50.03	-6.025	5.643	4404	1.136e+04	
V(act+,act-)	200f	200µ	50.03	-6.025	10.11	4071	1.229e+04	
V(act+,act-)	200f	205µ	50.03	-6.025	13.27	2456	2.037e+04	
V(act+,act-)	200f	210µ	50	-6.025	0.00865			notch shallower than 3 d
V(act+,act-)	200f	215µ	49.95	-6.025	0.002867			notch shallower than 3 d
V(act+,act-)	200f	220µ	49.89	-6.025	0.001741			notch shallower than 3 d
V(act+,act-)	200f	225µ	49.84	-6.025	0.001264			notch shallower than 3 d
V(act+,act-)	300f	175µ	50	-6.03	4.046	8680	5761	
V(act+,act-)	300f	180µ	50	-6.03	4.72	9532	5246	
V(act+,act-)	300f	185µ	50	-6.03	5.658	9877	5063	
V(act+,act-)	300f	190µ	50	-6.03	7.06	9789	5108	
V(act+,act-)	300f	195µ	50	-6.03	9.436	9277	5390	
V(act+,act-)	300f	200µ	50	-6.03	14.7	8273	6044	
V(act+,act-)	300f	205µ	50	-6.03	23.91	6581	7598	
V(act+,act-)	300f	210µ	50	-6.03	6.015	3316	1.508e+04	
V(act+,act-)	300f	215µ	49.99	-6.03	0.04976			notch shallower than 3 d
V(act+,act-)	300f	220µ	49.95	-6.03	0.01447			notch shallower than 3 d

E. Combined quality-factor table — all signals (page 2/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	300f	225µ	49.92	-6.03	0.008533			notch shallower than 3 d
V(act+,act-)	400f	175µ	49.98	-6.038	6.186	1.753e+04	2851	
V(act+,act-)	400f	180µ	49.98	-6.038	7.082	1.735e+04	2881	
V(act+,act-)	400f	185µ	49.98	-6.038	8.285	1.691e+04	2955	
V(act+,act-)	400f	190µ	49.98	-6.038	10.01	1.622e+04	3082	
V(act+,act-)	400f	195µ	49.98	-6.038	12.75	1.524e+04	3280	
V(act+,act-)	400f	200µ	49.98	-6.038	18.26	1.392e+04	3590	
V(act+,act-)	400f	205µ	49.98	-6.038	34.16	1.218e+04	4104	
V(act+,act-)	400f	210µ	49.98	-6.038	13.45	9786	5107	
V(act+,act-)	400f	215µ	49.98	-6.038	6.175	6039	8277	
V(act+,act-)	400f	220µ	49.98	-6.038	0.9208			notch shallower than 3 d
V(act+,act-)	400f	225µ	49.95	-6.038	0.06877			notch shallower than 3 d
V(act+,act-)	500f	175µ	49.96	-6.048	8.292	2.634e+04	1897	
V(act+,act-)	500f	180µ	49.96	-6.048	9.346	2.567e+04	1946	
V(act+,act-)	500f	185µ	49.96	-6.048	10.73	2.483e+04	2012	
V(act+,act-)	500f	190µ	49.96	-6.048	12.64	2.38e+04	2099	
V(act+,act-)	500f	195µ	49.96	-6.048	15.55	2.255e+04	2215	
V(act+,act-)	500f	200µ	49.96	-6.048	21.03	2.106e+04	2372	
V(act+,act-)	500f	205µ	49.96	-6.048	47.3	1.928e+04	2591	
V(act+,act-)	500f	210µ	49.96	-6.047	18.67	1.713e+04	2917	
V(act+,act-)	500f	215µ	49.96	-6.047	11.86	1.444e+04	3460	
V(act+,act-)	500f	220µ	49.96	-6.047	7.332	1.084e+04	4610	
V(act+,act-)	500f	225µ	49.96	-6.047	3.676	4522	1.105e+04	
V(act+,act-)	600f	175µ	49.93	-6.061	10.28	3.602e+04	1386	
V(act+,act-)	600f	180µ	49.93	-6.06	11.44	3.506e+04	1424	
V(act+,act-)	600f	185µ	49.93	-6.06	12.94	3.395e+04	1471	
V(act+,act-)	600f	190µ	49.93	-6.06	14.95	3.27e+04	1527	
V(act+,act-)	600f	195µ	49.93	-6.06	17.94	3.129e+04	1596	
V(act+,act-)	600f	200µ	49.93	-6.06	23.29	2.969e+04	1682	
V(act+,act-)	600f	205µ	49.93	-6.06	51.05	2.789e+04	1790	
V(act+,act-)	600f	210µ	49.93	-6.06	22.88	2.585e+04	1932	
V(act+,act-)	600f	215µ	49.93	-6.06	16.07	2.35e+04	2125	

E. Combined quality-factor table — all signals (page 3/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	600f	220µ	49.93	-6.06	11.83	2.075e+04	2406	
V(act+,act-)	600f	225µ	49.93	-6.059	8.604	1.741e+04	2867	
V(act+,act-)	700f	175µ	49.91	-6.076	12.12	4.685e+04	1065	
V(act+,act-)	700f	180µ	49.91	-6.076	13.36	4.568e+04	1093	
V(act+,act-)	700f	185µ	49.91	-6.075	14.92	4.439e+04	1124	
V(act+,act-)	700f	190µ	49.91	-6.075	17	4.3e+04	1161	
V(act+,act-)	700f	195µ	49.91	-6.075	20	4.147e+04	1203	
V(act+,act-)	700f	200µ	49.91	-6.075	25.17	3.981e+04	1254	
V(act+,act-)	700f	205µ	49.91	-6.075	43.47	3.8e+04	1313	
V(act+,act-)	700f	210µ	49.91	-6.075	26.49	3.602e+04	1385	
V(act+,act-)	700f	215µ	49.91	-6.075	19.49	3.384e+04	1475	
V(act+,act-)	700f	220µ	49.91	-6.074	15.34	3.143e+04	1588	
V(act+,act-)	700f	225µ	49.91	-6.074	12.29	2.872e+04	1738	
V(act+,act-)	800f	175µ	49.88	-6.094	13.81	5.893e+04	846.5	
V(act+,act-)	800f	180µ	49.88	-6.094	15.1	5.761e+04	865.9	
V(act+,act-)	800f	185µ	49.88	-6.094	16.7	5.62e+04	887.6	
V(act+,act-)	800f	190µ	49.88	-6.093	18.8	5.47e+04	911.9	
V(act+,act-)	800f	195µ	49.88	-6.093	21.79	5.311e+04	939.3	
V(act+,act-)	800f	200µ	49.88	-6.093	26.77	5.14e+04	970.4	
V(act+,act-)	800f	205µ	49.88	-6.093	41.51	4.959e+04	1006	
V(act+,act-)	800f	210µ	49.88	-6.092	29.72	4.765e+04	1047	
V(act+,act-)	800f	215µ	49.88	-6.092	22.39	4.557e+04	1095	
V(act+,act-)	800f	220µ	49.88	-6.092	18.25	4.334e+04	1151	
V(act+,act-)	800f	225µ	49.88	-6.092	15.28	4.093e+04	1219	